CHAPTER

6

COST, REVENUE AND PRODUCER'S EQUILIBRIUM

Course of Action
Maximum Time = 3:15 Hrs
Maximum Questions = 104

Learning Objectives

The students will learn:

- *The concept of cost and various types of cost.*
- *Meaning of Short Run Cost, their curves and calculations.*
- *Long run cost curves.*
- *Revenue, types along with curves and calculation.*
- *Producer's Equilibrium through both TC – TR approach and MC – MR approach.*

Topic-1
Cost Concepts **Page No. 93**
Topic-2
Revenue Concepts **Page No. 104**
Topic-3
Producer's Equilibrium **Page No. 110**

Topic-1 Cost Concepts

Concepts covered: Meaning and types of cost–a. Explicit and implicit cost, b. Private and social cost, c. Short run and long run cost. Short run cost– a. total cost, Total variable cost and total fixed cost, b. Average cost, average fixed cost and average variable cost, c. Marginal cost. Relationship between TC, TVC and TFC. Relationship between AC, AVC and AFC. Relationship between TC and MC.

Revision Notes

Scan to know more about this topic
Cost

- **Cost :** It refers to the expenditure incurred by a producer on the factor as well as non-factor inputs for a given amount of output of a commodity.
- **Cost Function :** A Cost Function shows the functional relationship between output and cost of production. It is given as :

$$C = f(Q)$$

- **Production Cost :** It refers to the expenditure incurred by a producer used in the process of production on factor as well as non-factor inputs.
- **Explicit Cost :** It refers to the actual money expenditure of a firm on purchasing goods or hiring factor services and non-factor inputs (like raw material, electricity, fuel, etc). Explicit Cost are those cash payments which firms make to outsiders for their services and goods.
- **Implicit Cost :** Implicit cost is the imputed or estimated value of inputs supplied by the owner of the firm himself. Implicit Cost are the costs of entrepreneur's own factors or resources.
- **Accounting Cost or Money Cost:** The money cost of producing a certain output of a commodity is the sum of all the payments to the factors of production engaged in the production of that commodity.
- **Opportunity Cost :** Opportunity cost is the cost of the next best alternative foregone.
- **Real Cost :** Real cost is the qualitative sum of all the efforts made by the owners of factor of production.

Scan to know more about this topic
Types of costs

- **Social Cost :** Social cost is the cost that the society has to bear for production of a commodity. Social cost = Private cost + External cost. Examples of such costs are pollution, etc.
- **Short Run Cost** : The cost incurred by a producer in the short run. Short Run cost are those in which some factors of production are fixed and others are variable. These are classified as Total Cost, Average Cost and Marginal Cost.
- **Total Fixed Costs or Supplementary Cost :** Fixed Cost are the sum total of expenditure incurred by the producer on the purchase of fixed factors of production. Fixed costs are those costs of production which do not change with a change in output.
- **Total Variable Costs :** Variable Costs are the expenditure incurred by the producer on the use of variable factors of production.

Cost

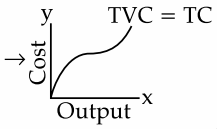
Cost Concepts

Basic Concepts

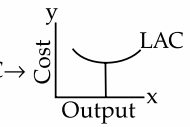
- **Social Cost**
  - Cost of purifying contaminated water by society.
- **Private Cost**
  - Cost of products
- **Real Cost**
  - Sacrifice + Inconvenience + discomfort + all other efforts involved in supplying factors of production by their owners.
- **Money Cost**
  - Cost of raw material, wages, rent etc.
- **Explicit Cost**
  - Money wages, rent, interest etc.
- **Implicit Cost**
  - Estimated rent of owners, own building, salary for the services of entrepreneurs etc.
- **Economic Cost**
  - Explicit cost + Implicit cost = Normal profits
- **Opportunity Cost**
  - Sacrifice made in terms of explicit or implicit cost for producing a given level of output.
    Opportunity cost = Explicit cost + Implicit cost
    - (Accounting cost) Payment to others for the purchase of inputs
    - Value of self owned inputs in thier next best alternative use

Long run costs

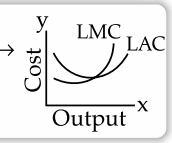
- **Total Cost**
  - TC = TVC →



- **Average Cost**
  - AC = AVC →



- **Marginal Cost**
  - MC = $TC_n – TC_{n–1}$ →

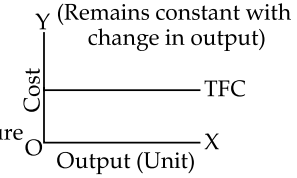


Short run Costs Schedule

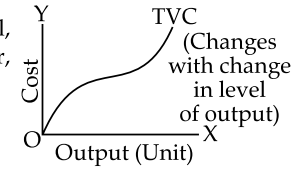
Output Unit	TFC	TVC	TC	AFC	AVC	AC	MC
0	12	0	12	-	-	-	-
1	12	9	21	12	9	21	9
2	12	16	28	6	8	14	7
3	12	18	30	4	6	10	2
4	12	22	34	3	5.5	8.5	4
5	12	30	42	2.4	6	8.4	8
6	12	40	52	2	6.6	8.6	10

Short run costs

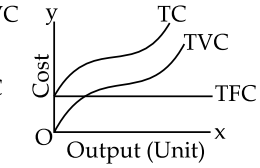
- **Total Fixed Cost**
  - Expenditure on machine, plant, license fee, expenditure etc.



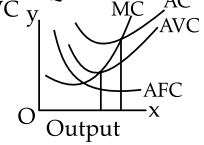
- **Total variable Cost**
  - Cost of raw material, wages of casual labour, etc.



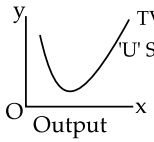
- **Total Cost**
  - Total Cost = TFC + TVC



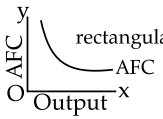
- **Total Cost**
  - Average cost ($AC = \frac{TC}{Q}$)
    AC = AFC + AVC



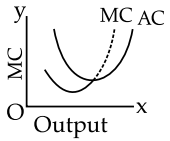
- **Average Variable Cost**
  - AVC = AC – AFC



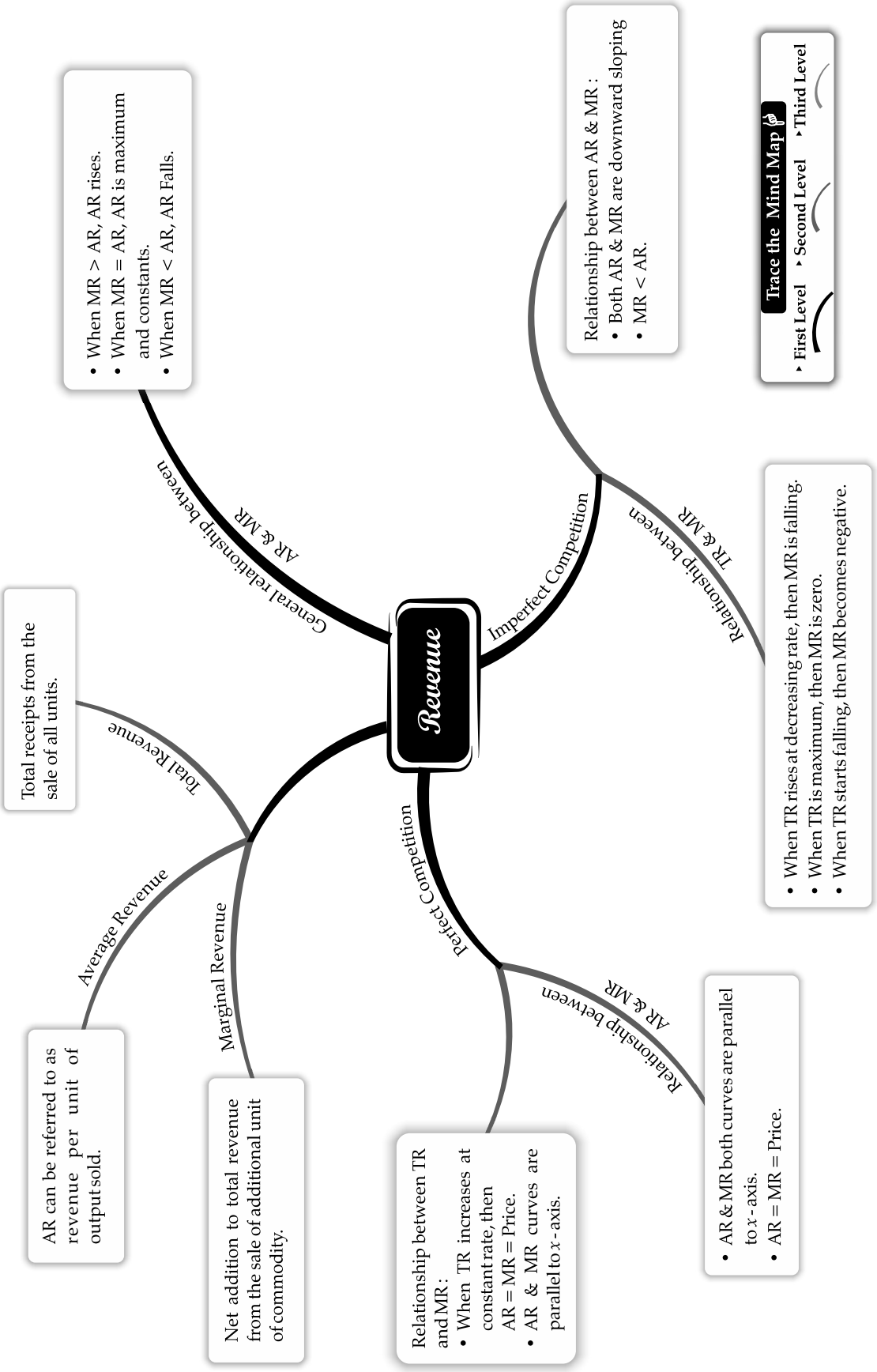
- **Average Fixed Cost**
  - AFC = TFC or AFC = AC – AVC

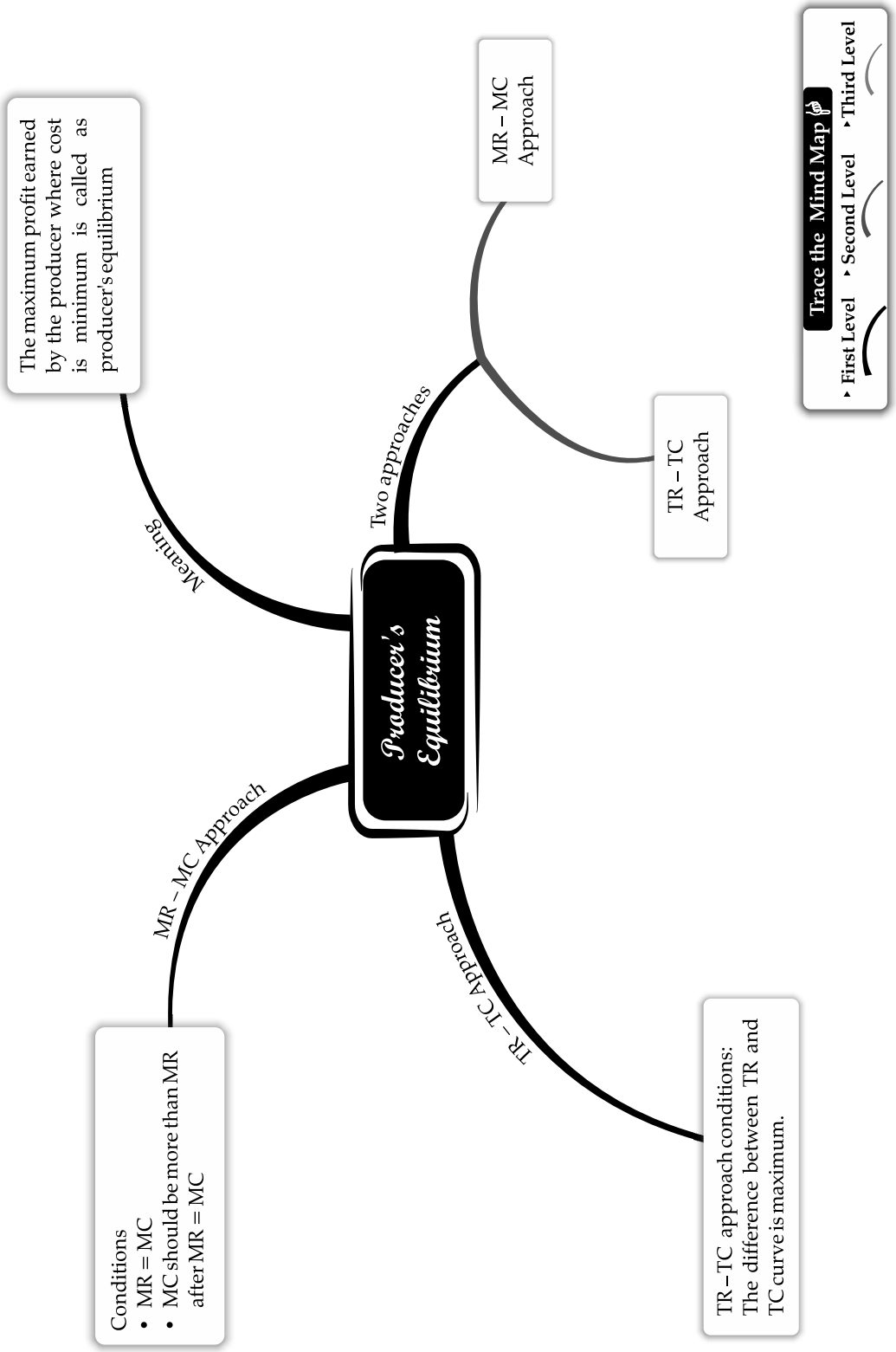


- **Marginal Cost**
  - $MC = \frac{\Delta TC}{\Delta Q}$ or $TC_n – Tc_{n–1}$



Trace the Mind Map
First Level · Second Level · Third Level


Revenue
Total Revenue
Total receipts from the sale of all units.
Average Revenue
AR can be referred to as revenue per unit of output sold.
Marginal Revenue
Net addition to total revenue from the sale of additional unit of commodity.
General relationship between AR & MR
• When MR > AR, AR rises.
• When MR = AR, AR is maximum and constants.
• When MR < AR, AR Falls.
Perfect Competition
Relationship between AR & MR
• AR & MR both curves are parallel to x - axis.
• AR = MR = Price.
Relationship between TR and MR :
• When TR increases at constant rate, then AR = MR = Price.
• AR & MR curves are parallel to x - axis.
Imperfect Competition
Relationship between TR & MR
• When TR rises at decreasing rate, then MR is falling.
• When TR is maximum, then MR is zero.
• When TR starts falling, then MR becomes negative.
Relationship between AR & MR :
• Both AR & MR are downward sloping
• MR < AR.
Trace the Mind Map
First Level
Second Level
Third Level


Producer's Equilibrium
Meaning
The maximum profit earned by the producer where cost is minimum is called as producer's equilibrium
Two approaches
MR – MC Approach
TR – TC Approach
MR – MC Approach
Conditions
• MR = MC
• MC should be more than MR after MR = MC
TR – TC Approach
TR – TC approach conditions: The difference between TR and TC curve is maximum.
Trace the Mind Map
First Level
Second Level
Third Level

- **Total Costs :** It is the total expenditure incurred by a firm on the factors of production required for the production of a commodity. It is a sum of total fixed cost and total variable cost.

$$TC = TFC + TVC$$

- **Average Costs :** Cost per unit of output is called Average Cost. It is obtained by dividing the total cost by the quantity of output.

$$AC = \frac{TC}{Q}$$

- **Average Fixed Cost (AFC)** : It is defined as the fixed cost of producing per unit of the commodity. It is obtained by dividing TFC by the level of output.

$$AFC = \frac{TFC}{Q}$$

- **Average Variable Cost (AVC) :** It is defined as the variable cost of producing per unit of commodity. It is obtained by dividing TVC by the level of output.

$$AVC = \frac{TVC}{Q}$$

- **Marginal Cost :** Marginal cost is the addition made to the total cost by the production of one more unit of a commodity.

$$MC = TC_n - TC_{n-1}$$

or $$MC = TVC_n - TVC_{n-1}$$

or $$MC = \frac{\Delta TC}{\Delta Q} = \frac{\Delta TVC}{\Delta Q}$$

- **Relationship between AC and MC :**
  - Both are derived from TC.
  - When AC falls, MC is lower than AC.
  - When AC rises, MC is greater than AC.
  - MC cuts AC at its minimum point.
- **Relationship between TC and MC** :
  - MC is the addition to total cost, when one more unit of output is produced. MC is calculated as;

$$MC_n = TC_n - TC_{n-1}$$

  - TC increases at an increasing rate when MC is increasing.
  - TC increases at a constant rate when MC is constant.
  - TC increases at a diminishing rate when MC is decreasing.
- **Relationship between TC, TFC and TVC**
  - TC is the sum total of TFC and TVC, i.e. TC = TFC + TVC
  - At zero level of output, TVC is zero, so TVC curve starts from origin.
  - At zero level of output, TFC is not zero, so TFC curve starts from a point F on the y-axis, above the origin.
  - At zero level of output, TC = TFC.
  - The difference between TC and TVC is TFC, which is constant. So, TC and TVC curves are parallel to each other.
  - Both TC and TVC curves rise at a diminishing rate and then at an increasing rate.
- **Long Run Costs :** The costs incurred by a producer in the long run. Since all factors of production are variable in long run, so there is no fixed costs in the long run.
- **Long Run Average Cost :** It is the sum total of the minimum levels of short run average costs at various levels of outputs.
- **Long run Marginal Cost :** Long Run Marginal Cost is the addition made to the total cost by the production of one more unit of a commodity when all factors of production are variable.

Example:

The following table shows the total cost schedule of a firm. Calculate the TVC, AFC, AVC, AC and MC schedules of the firm. [Q.25 of NCERT on page no.51]

Output	0	1	2	3	4	5	6
TC (₹)	10	30	45	55	70	90	120

Ans.

Output	**TC (₹)**	**TFC (₹) TC at 0 unit**	**TVC (₹) = TC – TFC**	**AFC (₹) = $\frac{TFC}{Q}$**	**AVC (₹) = $\frac{TVC}{Q}$**	**AC (₹) = $\frac{TC}{Q}$**	**MC (₹) = TCn – TCn – 1**
0	10	10	0	—	—	—	—
1	30	10	20	10	20	30	20
2	45	10	35	5	17.5	22.5	15
3	55	10	45	3.33	15	18.33	10
4	70	10	60	2.50	15	17.5	15
5	90	10	80	2.0	16	18	20
6	120	10	110	1.67	18.33	20	30

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. Why is the short run average cost curve U-shaped? [ISC Specimen Paper 2023 – 24]

Ans. The short run average cost curve is U– shaped because of the application of the law of variable proportions. That is, as the output increases the average cost decreases as total cost increases at a diminishing rate and then after reaching the minimum starts to increase as total cost increases at an increasing rate.

Q.2. What is meant by accounting cost? [ISC 2023]

Concept Applied	Types of Cost

Ans. The money cost of producing a certain output of a commodity is the sum of all the payments to the factors of production engaged in the production of that commodity.

Q.3. What is marginal cost of a firm? [OEB]

Ans. Marginal costs refer to addition to total cost when one more unit of output is produced.

$MC_n = TC_n - TC_{n-1}$

Marginal costs are based on cost of hiring/ purchasing factors of production that are variable, like – labour, materials, and equipment, for example – and not fixed costs the company will have whether it increases production or not.

Marginal cost is the change in total cost when one more unit of output is produced. However, when change in unit produced is more than one, then marginal cost can also be calculated as –

$$MC = \frac{\text{Change in Total Cost}}{\text{Change in units of Output}} = \frac{TC}{Q}$$

Q.4. Give the difference between total fixed cost and total variable cost? [OEB]

Ans. Total fixed cost can be defined as the amount spent on fixed factors of production whereas total variable cost can be defined as the amount spent on variable factors of production. Total fixed cost curve is parallel to X-axis whereas total variable cost curve is inverse S-shaped.

Q.5. State the difference between Average fixed Cost and Average Variable Cost? [OEB]

Q.6. State with reason whether the given statement is true or false: [OEB]
"Average cost is sum total of Average Fixed Cost and Average Variable Cost."

Concept Applied	Short Run Cost

Ans. True

Reason: Average cost is the sum total of Average Fixed Cost and Average Variable Cost. Average fixed cost is calculated by dividing total fixed cost by output and Average Variable Cost is calculated by dividing Total variable cost by output.

Q.7. "MC is the difference between total cost or total variable cost when an additional unit of output is produced." [OEB]

Ans. True

Reason: The given statement is true as marginal cost is the difference between total cost or total variable cost. The net addition to total cost or total variable cost with an additional unit of output is called as marginal cost.

Q.8. "The difference between average cost and average variable cost goes on rising." [OEB]

Ans. False

Reason: The given statement is false because the difference between average cost and average variable cost is average fixed cost which goes on falling as the output is increased.

Q.9. "The marginal cost curve cuts the average cost curve from above." [OEB]

Ans. False

Reason: The marginal cost curve is the addition to total variable cost when an additional unit of output is produced. Average Cost is the per unit cost of total output. Hence when MC cuts the AC at its minimum and from below.

2 Marks Questions

Q.1. Calculate total variable cost and marginal cost from the data given below. [ISC 2023]

Output (units)	0	1	2	3
Total cost	40	60	78	97

Ans.

Output (Units)	**Total Cost (TC)**	**Total Fixed Cost (TFC) (Cost at 0 unit)**	**Total Variable Cost (TVC) TVC = TC – TFC**	**Marginal Cost (MC) MC = TCn – TCn – 1**
0	40	40	0	–
1	60	40	20	20
2	78	40	38	18
3	97	40	57	19

This question is for practice and its solution is available at the end of the chapter

Q.2. Differentiate between accounting cost and opportunity cost. **[ISC 2019]**

OR

Give one difference between accounting cost and opportunity cost. **[ISC 2015]**

Concept Applied Types of Cost

Ans. The difference between accounting cost and opportunity cost are:

Accounting cost	Opportunity cost
Accounting cost is the **money payment** for hiring or buying factors of production. i.e., wages paid or money spent for buying raw materials.	Opportunity cost is the **probable loss of income** opportunities of the factors of production from their next best alternative uses. /Opportunity cost of producing any good is the **next best alternative good** that is given up producing this good.
It is an **explicit** cost.	It is an **implicit** cost.

[ISC Marking Scheme, 2019]

Examiner's Comments

- Most of the candidates neither used the term next best alternative nor did they explain opportunity cost in the correct manner. While explaining accounting cost, several candidates did not write that it was the cost of hiring various factors of production.

Answering Tip

- While explaining opportunity cost, explain the meaning of next best alternative with the help of examples.

Q.3. Define implicit cost. How is it different from explicit cost? **[ISC 2018]**

Concept Applied Types of Cost

Ans. Implicit cost refers to the imputed or estimated value of inputs owned by the firm used in its own production unit. Explicit cost refers to the money payment made by the firm to owners of factor services required for production is known as explicit cost. **[ISC Marking Scheme, 2018]**

Examiner's Comments

- Several candidates did not write that implicit cost is the estimated value of inputs owned or that is the imputed/estimated value of the factor owned by the producer and used in his own production process.

Answering Tip

- Learn the difference between implicit cost and explicit cost clearly.

Q.4. Explain the shape of MC curve. **[ISC 2017]**

Ans. It is a U shaped curve. It is due to law of variable proportions. Cost fall due to increasing returns to factor and then rises due to diminishing returns to factor. **[ISC Marking Scheme, 2017]**

Examiner's Comments

- Several candidates did not write that the U shape of MC curve is because of the law of variable proportions. A few candidates did not explain the relation between the law and the shape of the MC curve.

Answering Tip

- Understand the shape of the MC curve with the law of variable proportions.

Q.5. Explain the meaning of social cost with the help of an example. **[ISC 2016]**

Concept Applied Types of Cost

3 Marks Questions

Q.1. Calculate AC, AVC and AFC from the following: **[ISC Specimen Paper 2023-24]**

TFC is ₹15 at all levels of output.

Output (Units)	1	2	3	4
TVC	10	19	29	40

This question is for practice and its solution is available at the end of the chapter

Ans.

Output (Units)	TVC	TFC	TC = TFC + TVC	AC = $\frac{TC}{Q}$	AVC = $\frac{TVC}{Q}$	AFC = $\frac{TFC}{Q}$
1	10	15	25	25	10	15
2	19	15	34	17	9.5	7.5
3	29	15	44	14.67	9.67	5
4	40	15	55	13.75	10	3.75

Q.2. Give reasons for the following: [ISC 2020]

(i) TC and TVC curves do not start from the same point.

(ii) ATC and AVC curves do not touch each other.

Ans. (i) Total Cost = Total Variable Cost + Total Fixed Cost.

When we see the Total Variable cost it is 0 at 0 level of production and then starts increasing once the production starts. Thus, TVC starts from origin as at zero level of output as there is no variable cost incurred. But even at 0 level of output the firm incurs fixed cost, which remains constant. Thus, at 0 level of output, TC will still have the total fixed cost, so TC starts from a point where TFC also starts and not from the same point as TVC.

(ii) AVC and AC curve tend to intersect but never intersect each other as the value of AFC continuously declines as output rises and even tends to become zero but never becomes zero.

Examiner's Comments

- Many candidates wrote, TC=TFC+TVC as the reason. They did not write the reason of TC and TVC curves starting from different points. A few candidates wrote that TC starts from the point where TFC starts but hardly any candidate mentioned the starting point of TVC, thereby making the answer incomplete.
- Majority of the candidates did not write AFC is not equal to Zero which is the main logic behind the statement.

Answering Tip

- The students need to understand that TC starts from the point where TFC starts because at zero output TVC is zero and TC = TFC. Focus on the fact that AFC can never be zero because TFC is always positive and AFC= TFC/Q. The relationship with cost curves has to be understood with relevant examples of types of cost in the production process. Relationship between TC, TFC, TVC and AC, AFC and AVC should be learnt in the form of a table. The gap between AC and AVC should be shown to be decreasing and the reason for the same should be learnt and understood.

Q.3. Study the cost function of a firm given below:

Output (Units)	0	1	2	3
TC (₹)	30	90	110	120

Calculate: [ISC 2019]

(i) AFC

(ii) AC

(iii) MC

Ans.

Output	TC	TFC (TC at 0 output level)	AFC = $\frac{TFC}{Output}$	AC = $\frac{TC}{Output}$	MC = TCn – TCn-1
0	30	30	-	-	-
1	90	30	30	90	60
2	110	30	15	55	20
3	120	30	10	40	10

[ISC Marking Scheme, 2019]

Examiner's Comment

- Majority of the candidates answered this question correctly. However, some candidates did not answer all the three values, missing out either AFC or AC. A few candidates were unclear of the calculations at zero level of output.

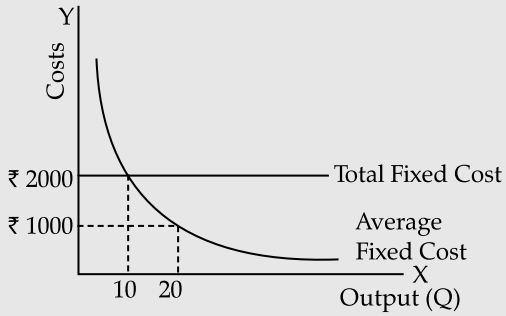
Answering Tips

- Practise adequately on questions based on cost function with proper formulas.
- The students need to focus on the fact that TC at zero level of output is TFC.

Q.4. Explain the shapes of total fixed cost curve and average fixed cost curve. Give one reason each, to justify the shape of the two curves. [ISC 2018]

Concept Applied Short Run Cost

Ans. Fixed costs (FC) are independent of output and must be paid out even if the production stops. AFC = Total Fixed Costs (FC)/Output (Q).



Total fixed cost is horizontal straight line running parallel to the X axis as it is fixed and does not change with output.

Average fixed cost is a rectangular hyperbola, asymptotic curve. It is downward sloping and does not touch the X axis or the Y axis. AFC is TFC / Q hence, as output increases AFC decreases but does not become zero as TFC is never zero.

[ISC Marking Scheme, 2018]

Examiner's Comment

- Several candidates were unable to write the proper reasons of a horizontal total fixed cost curve and asymptotic average fixed cost curve. Some candidates were unable to draw the correct shape of the AFC curve.

Answering Tip

- Give adequate practice in drawing the shape of total fixed cost curve and average fixed cost curve.

Q.5. Differentiate between real cost and money cost with the help of examples. [ISC 2017]

Concept Applied Types of Cost

Q.6. The difference between AC curve and AVC curve decreases with increase in output but the two curves never touch each other. Justify the statement with the help of a diagram. [ISC 2017]

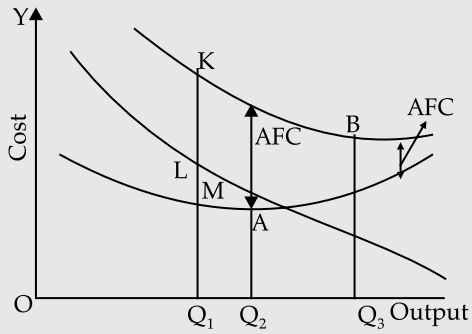
Ans. Difference between AC and AVC:

We know that, ATC – AVC = AFC

And AFC = TFC / Output.

Therefore, as output increases, AFC falls. So, the difference between ATC and AVC decreases with increase in output.

ATC and AVC can never be equal at any level of output because AFC can never be zero as TFC is positive.



[ISC Marking Scheme, 2017]

Examiner's Comment

- Most of the candidates did not answer this part correctly.

Answering Tip

- Students should practice diagrams.

Q.7. Classify the following into fixed cost and variable cost giving a reason for your answer: [ISC 2016]

(i) Expenses incurred on raw material

(ii) Interest on capital

(iii) Salaries to permanent employees

Ans. Classification of fixed cost and variable cost:

(i) Expenses on raw materials is a variable cost as more raw materials will be needed if the output increases, increasing the cost.

(ii) Interest on capital is a fixed cost as it has to be paid at the fixed agreed rate even if there is no production taking place.

(iii) Salaries to permanent employees is a fixed cost as the permanent employees have to be paid the salary even if there is zero level of production.

This question is for practice and its solution is available at the end of the chapter

Examiner's Comments

- While most candidates were able to classify the expenses correctly, many lost marks in giving the reason. Very poor explanation was given by most candidates.
- Candidates were able to identify correctly. In a few cases, it was classified as variable cost.
- Candidates made mistakes in giving explanation for example: "permanent employees are permanent and so salaries is fixed"

Answering Tips

- Understand the behaviour of factors in the short run with reference to the change in output and hence the behaviour of cost.
- Analyse that interest on capital is a fixed cost as the interest has to be paid even though there is no production or output in zero.
- Relate that salaries of permanent staff are fixed costs as this expenditure has to be incurred by the producers even though, there is no production.

Q.8. Explain the shape of the following: [ISC 2016]

(i) Total fixed cost curve.

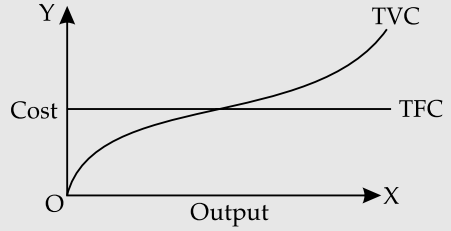
(ii) Total variable cost curve.

Ans. (i) Total fixed cost curve:

The total fixed cost is a straight line parallel to the x – axis.

(ii) Total variable cost curve:

Total variable cost is a concave downward up-to a certain level of output and subsequently it is concave upward.



[ISC Marking Scheme, 2016]

Examiner's Comments

- Most of the candidates were able to draw and explain the shape of TFC. Some drew correctly but were not able to explain the shape.
- Most of the candidates were not able to draw the TVC curve correctly. They drew an upward sloping curve – Straight line form the origin.

Answering Tips

- Learn the shape of TFC with fixed factor which remains constant in the short run.
- Refer to the law of variable proportions while learning the shape of TVC. It should be concave downwards and then concave upward.

Q.9. The cost function of a firm is given below:

Output	0	1	2	3	4
Total cost ₹	100	250	370	550	740

Calculate: [ISC 2015]

(i) AFC (ii) AVC (iii) MC

Ans.

Output	0	1	2	3	4
TC ₹	100	250	370	550	740
TFC (TC at 0 Unit output)	100	100	100	100	100
TVC = TC – TFC	0	150	270	450	640
AVC = TVC ÷ Output	–	150	135	150	160
AFC = TFC ÷ Output	–	100	50	33.33	25
MC = TCn – TCn – 1	–	150	120	180	190

[ISC Marking Scheme, 2015]

Examiner's Comment

- Some candidates were confused with '0' and infinite. However, a large number of candidates were able to attempt this part correctly.

Answering Tip

- It should be kept in mind that TFC = TC at zero level of output.

4 Marks Questions

Q.1. Define average variable cost, average total cost and marginal cost. State the relationship between AVC, ATC and MC. [OEB]

Ans. Average variable Cost can be defined as the total variable cost divided by output. Average total Cost can be defined as total cost divided by output. Marginal Cost can be defined as the net addition to total cost when an additional unit of output is produced.

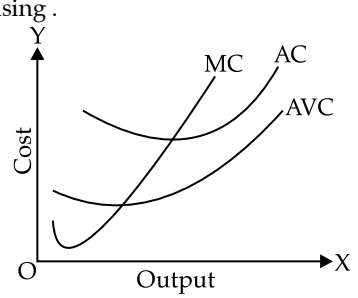
Relationship between AVC, ATC and MC is as follows:

(i) MC cuts AVC and ATC at their minimum points.

(ii) The shape of both AVC and ATC is U-shaped.

(iii) AVC lies below ATC and MC first cuts AVC then ATC.

(iv) The difference between ATC and AVC is AFC which goes on falling as TFC divided by units of output is increasing .

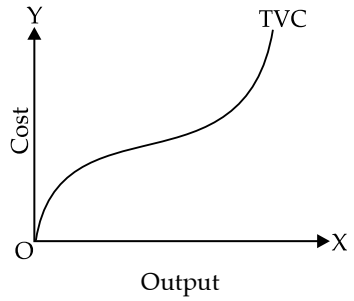


Q.2. Give the meaning of cost. State the behaviour of (a) total variable cost, and (b) average fixed cost as output rises. [OEB]

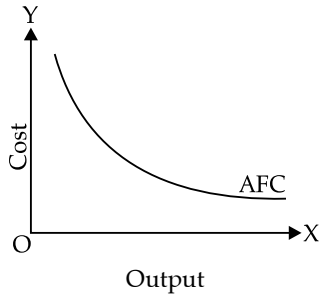
Q.3. (a) Give the reason for inverse S-shape of total variable cost.

(b) Define average fixed cost. Why is AFC rectangular hyperbola? Describe with diagram.

Ans. (a) The inverse S-shape of total variable cost of production is due to the reason that first total variable cost increases at decreasing rate and then increases at increasing rate.



(b) Average fixed cost can be defined as total fixed cost divided by output. As output rises, average fixed cost falls continuously as total fixed cost is divided by output as output rises, average fixed cost also falls.



6 Marks Questions

Q.1. What do you mean by economic cost? With the help of a diagram explain the relationship between the Total cost, Total Fixed Cost and Total Variable Cost. [OEB]

Concept Applied	Types of cost and Short-Run cost

Ans. The economic cost is the algebraic aggregate of all explicit and implicit costs incurred by the producer in the production process. The explicit costs are those cash payments which firms make to outsiders for their services and goods, whereas, the implicit costs are the costs of entrepreneur's own factors or resources.

Relationship between TC, TFC and TVC :

(i) TC is the sum total of TFC and TVC, *i.e.*

TC = TFC + TVC

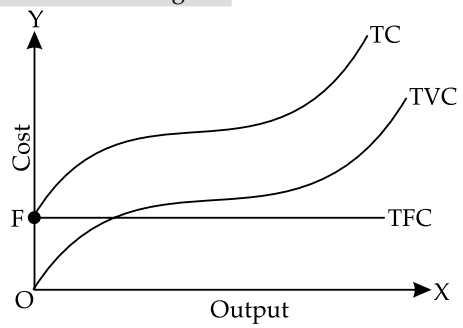
(ii) At zero level of output, TVC is zero, so TVC curve starts from origin

(iii) At zero level of output, TFC is not zero, so TFC curve starts from a point F on the y-axis, above the origin.

(iv) At zero level of output, TC = TFC, so TC curve starts at the same point F where the TFC curve starts.

(v) The difference between TC and TVC is TFC, which is constant. So, TC and TVC curves are parallel to each other.

(vi) Both TC and TVC curves rise at a diminishing rate and then at increasing rate.



Q.2. The cost function of a firm is given below :

Output (units)	**0**	**1**	**2**	**3**	**4**
Total Cost (₹)	**120**	**160**	**200**	**222**	**232**

Calculate: (i) Total Fixed Cost **(ii) Total Variable Cost**

(iii) Average Fixed Cost **(iv) Average Variable Cost**

(v) Marginal Cost **[OEB]**

This question is for practice and its solution is available at the end of the chapter

Ans.

Output (units)	Total Cost (₹)	Total Fixed Cost (₹)	Total Variable Cost (₹)	Average Fixed Cost (₹)	Average Variable Cost (₹)	Marginal Cost (₹)
0	120	120	0	-	-	-
1	160	120	40	120	40	40
2	200	120	80	60	40	40
3	222	120	102	40	34	22
4	232	120	112	30	28	10

Q.3. Explain the relationship between Average Cost and Marginal Cost. **[OEB]**

Topic-2 Revenue Concepts

Concepts covered: 1. Meaning of revenue, 2. Types of revenue–a. Total revenue, b. Average revenue, c. Marginal Revenue, 3. Relationship between TR, AR and MR under, a. Perfect competition, b. Imperfect Competition

Revision Notes

- **Revenue :** Amount received by the producer upon sale of the output during a given period. Revenue includes cost incurred by the producer and the profit earned.
- **Total Revenue :** The total money receipts of a firm from the sale of a given output is called Total Revenue.

Scan to know more about this topic

Revenue Concept

- **Average Revenue (AR) :** Average revenue is the per unit revenue received from the sale of one unit of a commodity.

$$AR = \frac{TR}{Q}$$

- Average revenue curve and demand curve are one and same thing.
- Average revenue is also called firm's price line.
- **AR Curve is Firm's Demand Curve :** Firm's demand curve is a curve showing relationship between price of the products and its quantity demanded in the market.
- **Marginal Revenue :** Marginal revenue is the change in total revenue which results from the sale of one more or one less unit of output.

$$MR = TR_n - TR_{n-1}$$

Or, $$MR = \frac{\Delta TR}{\Delta Q}$$

- **Negative (MR) :** It is possible only when price is declining under monopoly or monopolistic competition. It is not possible in case of perfect competition, where price remains constant for a firm.
- **Revenue Curves in Different Markets :**
  - In perfect competition, AR is a horizontal line parallel to 'X' axis. It is equal to MR and TR curve is a straight positively sloping line from the origin and TR increases in same proportion as increase in output sold.
  - The area below the price line is total revenue in perfect competition.
  - In monopoly and monopolistic competition, AR and MR both are downward sloping and MR is always below AR. The main difference in these two markets is that in monopolistic competition AR and MR curve is more elastic than monopoly.
- **Relationship between TR and MR :**
  - Initially or at first unit TR = MR
  - When TR increases at increasing rate, then MR also increases.
  - When TR increases at constant rate, MR is also constant.
  - When TR increases at a diminishing rate, then MR declines.
  - When TR is maximum, MR is zero.
  - When TR declines, MR is negative.
- **Relationship between AR and MR :**
  - If AR is constant, AR = MR
  - If AR is diminishing, AR > MR
  - MR can be negative, but not AR. AR is always positive.
- **AR Curve is a Horizontal Straight Line under Perfect Competition :** A firm under perfect competition is a price taker. It cannot influence/change the market price, implying a constant AR for a firm corresponding to all levels of output.

This question is for practice and its solution is available at the end of the chapter

- **AR Curve Slopes Downwards under Conditions of Monopoly and Monopolistic Competition :** Under monopoly and monopolistic competition, more of the commodity can be sold only at a lower price. This implies an inverse relationship between price of the commodity and demand for the firm's output. Hence, a downward sloping firm's demand curve.

Example:

Calculate the AR and MR from the following data:

Output	1	2	3	4	5	6	7
TR	100	180	240	280	300	300	280

Units of Output (Q)	TR (₹)	AR = $\frac{TR}{Q}$	MR = ∑TR or TRn – TRn–1
1	100	100	100
2	180	90	80
3	240	80	60
4	280	70	40
5	300	60	20
6	300	50	0
7	280	40	–20

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. Define Marginal revenue. [OEB]

Ans. Marginal revenue cam be defined as the increase in total revenue when an additional unit of output is sold.

Q.2. Give the meaning of average revenue. [OEB]

Q.3. State the difference between the shape of MR curve in perfect competition and imperfect competition. [OEB]

Ans. The shape of MR curve in perfect competition is parallel to x-axis and the shape of MR curve in imperfect competition is downward sloping.

Q.4. Give the difference between the shape of demand curve under perfect competition and imperfect competition. [OEB]

Ans. The demand curve under perfect competition is parallel to x-axis and the demand curve under imperfect competition is downward sloping demand curve.

Q.5. Under imperfect competition, which curve among AR and MR is more steeper? [OEB]

Ans. Under imperfect competition, MR curve is more steeper than AR curve. Because Marginal revenue falls at more speed than average revenue.

Q.6. State the relationship between average revenue and marginal revenue under perfect competition. [OEB]

Ans. Under perfect competition, both average revenue and marginal revenue are straight line parallel to x-axis.

Q.7. State giving reason whether the following statement is true or false: [OEB]
"When TR increases at a constant rate, then MR is falling."

Ans. False

Reason: The given statement is false because when TR increases at a constant rate then MR remains same.

Q.8. "When TR is increasing at decreasing rate, then MR is rising." [OEB]

Ans. False

Reason: The given statement is false because when TR increases at decreasing rate then MR falls.

Q.9. "Under imperfect competition, shape of AR and MR curve is downward sloping." [OEB]

Ans. True

Reason: Under imperfect competition, Shape of AR and MR curve is downward sloping because more output can be sold in imperfect competition by reducing the price, hence both AR and MR curves are downward sloping.

Q.10. Under perfect competition, AR curve is downward sloping. [OEB]

Ans. False

Reason: The given statement is false. Under perfect competition, AR curve is straight line parallel to x-axis because price remains constant as more unit of output is sold.

Q.11. Under imperfect competition, when TR is falling but remains positive then MR falls and then it becomes negative. [OEB]

Ans. False

Reason: TR will be falling and will remain positive if MR is falling and is positive under imperfect competition.

2 Marks Questions

Q.1. "Concept of Average Revenue is also called price." Comment. [OEB]

Ans. Concept of Average Revenue is called price.

Given $AR = \frac{TR}{Q}$

Since, $TR = P \times Q$

$\therefore AR = \frac{P \times Q}{Q}$

$\therefore AR = P$

Q.2 Name the market where average revenue is equal to marginal revenue. Give a reason for your answer. [ISC 2015]

This question is for practice and its solution is available at the end of the chapter

Q.3. Complete the following table: [OEB]

Output sold	TR	AR	MR
0	0	__	__
1	10	__	10
2	__	10	10
3	30	10	__

Ans.

Output sold	TR	AR	MR
0	0	-	-
1	10	10	10
2	20	10	10
3	30	10	10

Q.4. Complete the following table: [OEB]

Output sold	TR	AR	MR
0	0	__	__
1	__	20	20
2	35	__	15
3	45	15	__

Ans.

Output sold	TR	AR	MR
0	0	__	__
1	20	20	20
2	35	17.5	15
3	45	15	10

Q.5. If TR at 5 units of output is ₹70 and at 10 units of output is ₹100 then calculate the average revenue? [OEB]

Output	Total revenue	Marginal revenue
5	70	__
10	100	30/5 = 6

Ans.

Output	Total revenue	Marginal revenue	Average Revenue
5	70	__	16
10	100	30/5 = 6	10

Q.6. If AR at 3 units of output is sold is ₹7 and at 6 units of output sold is ₹6, then calculate the MR at 6 units of output sold? [OEB]

Output sold	AR
3	7
6	6

Ans.

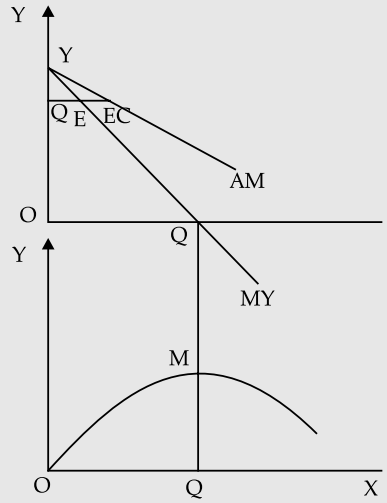
Output sold	AR	TR	MR
3	7	21	-
6	6	36	15

3 Marks Questions

Q.1. Explain with the help of a diagram, the relationship between MR and TR under imperfect competition. [ISC 2020]

Ans. The relationship between marginal revenue and total revenue can be explained as follows:

- When MR is **positive**, TR **increases** with an increase in output.
- When MR is **zero**, TR is **maximum**
- When MR becomes **negative**; TR **falls** with an increase in output.



[ISC Marking Scheme, 2020]

Examiner's Comments

The common errors made by the candidates were:

- Having drawn the incorrect diagram for MR and TR with no correspondence being established between maximum point of TR curve and point of intersection of MR curve with the horizontal or quantity axis, i.e., zero MR.
- Drawing diagrams with TR curve and MR curve side by side and hence no correspondence.
- Giving an incomplete presentation of one of the relations between TR and MR by only writing that when TR increases (at diminishing rate), MR decreases. They should have completed the sentence with incorporation of 'MR remains positive'.
- Having written MR in place of TR and vice-versa.

Answering Tip

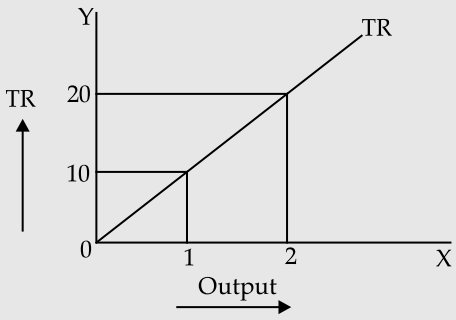
- The students need to remember that since a diagram is the basis of most of the answers in microeconomics, it is imperative for it to be meaningful. Also that even when MR is falling throughout, its sign (positive/zero/negative) varies with the behaviour of TR. Adequate practice should be done by the students in drawing the diagrams.

Q.2. With the help of a diagram, explain the relationship between AR and MR of a firm under imperfect competition. [ISC 2019]

Q.3. Draw the TR and AR curves under perfect competition with the help of a schedule. [ISC 2016]

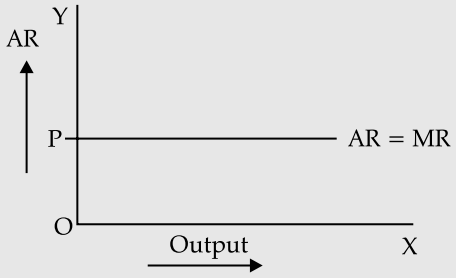
Ans. TR and AR curves under perfect competition:

Shape of TR and AR in perfect market



Price is constant in nature in perfect market. So, **TR curve starts from origin** and goes upward. TR increases **in direct proportion to output** due to constant price. It increases at constant rate.

Marginal and Average revenue



AR = MR and the curve is horizontal to x axis at level of ruling price OP.

Under perfect market the firm is not required to reduce the price to sell more. So, AR is constant at all level of output. If price remains same, when more units of a product are sold then MR = AR = P

Units of output(Q)	Price (in ₹)	TR = Q × Price	AR = TR/Q
1	5	5	5
2	5	10	5
3	5	15	5
4	5	20	5
5	5	25	5
6	5	30	5

[ISC Marking Scheme, 2016]

Examiner's Comment

- Most of the candidates were able to draw the TR and AR curves. Most errors were found in making the schedule.

Answering Tip

- Explain that in perfect competition, product is homogeneous and therefore, the price remains constant. TR cannot be obtained without price and output TR= P × Q. And AR= Price.

Q.4. When by reducing the price, the firm sells more quantity of a goods. What will be the behaviour of marginal revenue in that case and would it be greater than or less than or equal to average revenue? [OEB]

Concept Applied Revenue Concept

Ans. Under imperfect competition, when the firm sells more quantity only by lowering the price, then both marginal revenue and average revenue are downward sloping and marginal revenue is less than average revenue.

Q.5. "Marginal Revenue is below the Average Revenue curve under imperfect competition." Explain. [OEB]

Ans. Under imperfect competition, a producer can increase sales by decreasing the price. As a result, each additional unit is sold at the price lower than the price earned for the previous unit that is why MR curve is below AR curve under imperfect competition.

4 Marks Questions

Q.1. Complete the following table: [OEB]

Output (Units)	Average Revenue (AR in ₹)	Marginal Revenue (MR in ₹)	Total Revenue (TR in ₹)
1	—	15	—
2	—	—	26
3	11	—	—
4	—	3	—

Ans.

Output (Units)	Average Revenue (AR in ₹)	Marginal Revenue (MR in ₹)	Total Revenue (TR in ₹)
1	15	15	15
2	13	11	26
3	11	7	33
4	9	3	36

This question is for practice and its solution is available at the end of the chapter

Q.2. Complete the following table: **[OEB]**

Price (₹)	Output (Units)	Total Revenue (TR in ₹)	Marginal Revenue (MR in ₹)
10	1	—	—
—	2	14	—
—	3	—	1
—	4	12	—

Ans. Price = TR/ Q
Output = TR/ Price or TR/ AR
TR = AR × Q or SMR
MRn = TRn – TRn–1
AR = TR/ Q

Price (₹)	Output (Units)	Total Revenue (TR in ₹)	Marginal Revenue (MR in ₹)
10	1	10	10
7	2	14	4
5	3	15	1
3	4	12	(–) 3

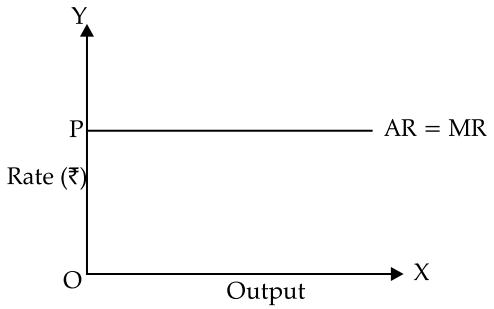
Q.3. Define Revenue. State the relation between marginal revenue and average revenue. **[OEB]**

Concept Applied Revenue Concept

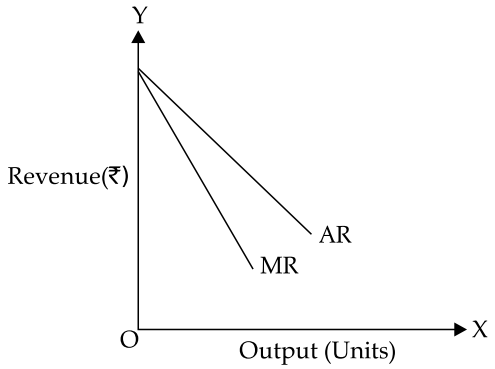
Ans. Revenue: The revenue of a firm is its sales, receipts or income.

Relation Between Marginal Revenue and Average Revenue:

(i) **Under Perfect Competition:** Under perfect competition, **firms are price taker** and firms can sell the desired output at this price. Therefore, AR = MR and both curves are **horizontal straight line parallel to X-axis.**



(ii) **Under Monopoly:** Monopoly is such a market situation in which there is a single seller having complete control over the supply of the commodity. The average revenue curve is **downward sloping** and its corresponding **marginal revenue curve lies below it.** The reason for downward sloping AR and MR curves is that, the monopolist can increase his sales by lowering the price. Marginal revenue also falls **but the rate of fall in MR is greater than in AR**. This is due to the fact that not only the additional units will be sold at lowered price but the whole units are to be sold at lower price.



Q.4. Complete the following table: **[OEB]**

Price (₹)	Output (Units)	Total Revenue (TR in ₹)	Marginal Revenue (MR in ₹)
—	1	6	—
4	—	—	2
—	3	6	—
1	—	—	(–) 2

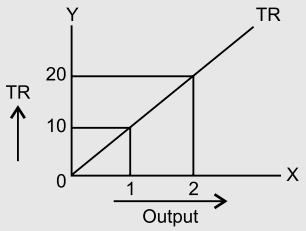
Ans.

Price (₹) $P = \frac{TR}{Q}$	Output (Units) $Q = \frac{TR}{P \text{ or } AR}$	Total Revenue (TR in ₹) TR = AR × Q Or TR = ∑MR	Marginal Revenue (MR in ₹) MR = $TR_n - TR_{n-1}$
6	1	6	6
4	2	8	2
2	3	6	(–) 2
1	4	4	(–) 2

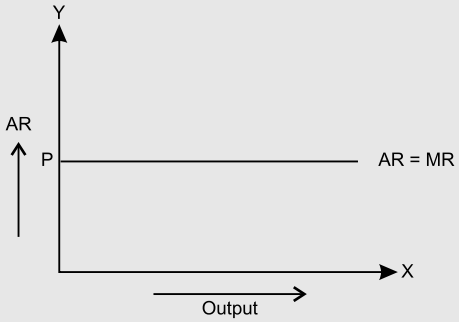
6 Marks Questions

Q.1. Draw TR and AR curves under perfect competition with the help of a schedule. **[ISC 2016]**

Ans. TR and AR curves under perfect competition :
Shape of TR and AR in perfect market



Price is constant in nature in perfect market. So, TR curve starts from origin and goes upward. TR increases in direct proportion to output due to constant price. It increases at constant rate.



AR = MR and the curve is horizontal to x axis at level of ruling price OP.

Under perfect market, the firm is not required to reduce the price to sell more. So, AR is constant at all levels of output. If price remains same, when more units of a product are sold, then, MR = AR = P.

[ISC Marking Scheme 2016]

Examiner's Comment

- Most of the candidates were able to draw the TR and AR curves. Most errors were found in making the schedule.

Answering Tip

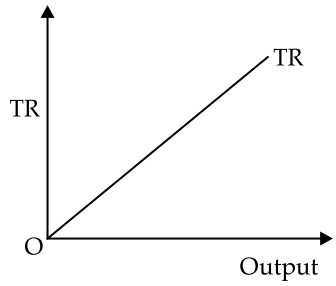
- Students must properly learn to make the schedule.

Q.2. How is the Average Revenue and Marginal Revenue under both perfect and imperfect Competition?

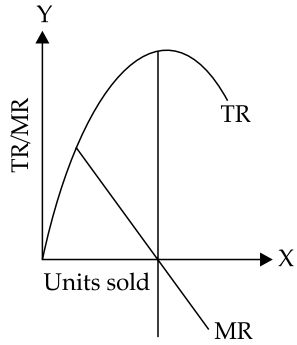
[OEB]

Q.3. With the help of suitable diagrams, differentiate between the shapes of Total Revenue Curve under both Perfect and Imperfect Competitions.

Ans. Under Perfect Competition, price or AR is constant as well as MR is constant. The rate of change in TR, that is, MR is constant. As a result, TR curve is a positively sloped straight line starting from origin.



Under Imperfect competition, price or AR is lowered to sell more. As a result, MR is falling. The rate of change in TR, that is MR, initially decreases at a diminishing rate and then becomes negative. As a result, TR initially increases at a diminishing rate but soon it starts falling.

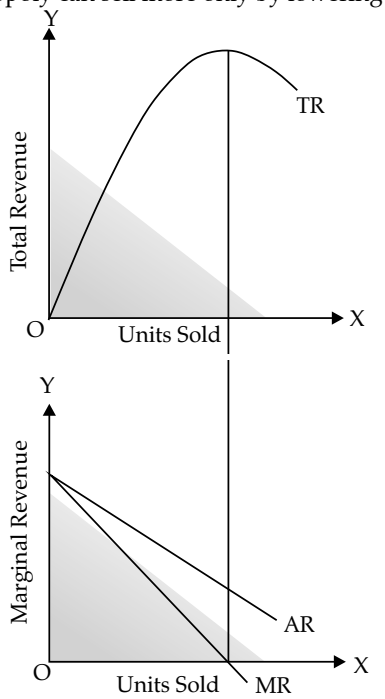


Q. 4. Explain with the help of a diagram, the relationship between MR and TR under imperfect competition.

[OEB]

Concept Applied	Revenue Cost

Ans. A firm under imperfect competition such as under monopoly can sell more only by lowering its price.



Relationship between TR and MR

This question is for practice and its solution is available at the end of the chapter

Under imperfect competition conditions, total revenue increases at a diminishing rate. It becomes maximum and then begins to decline. Under Imperfect competition, price or AR is lowered to sell more. As a result, MR is falling. The rate of change in TR, that is MR, initially decreases at a diminishing rate and then becomes negative. As a result, TR initially increases at a diminishing rate but soon it starts falling.

Topic-3 Producer's Equilibrium

Concepts covered: 1. Concept of producer's equilibrium, 2. Conditions of producer's equilibrium, MC-MR Approach, TC-TR Approach

Revision Notes

- **Producer's or Firm's Equilibrium :** A producer is said to be in equilibrium when he maximises his profit or minimises his losses. A producer is said to be in equilibrium when he produces that level of output at which his profits are maximum. Producer's equilibrium is also known as profit maximisation situation.

Scan to know more about this topic

Producer's Equilibrium

- **Profit :** The difference between TR and TC is profit.

$$\text{Profit} = TR - TC$$

For example, Total revenue= 2500; total cost= 1600; thus profit= 2500-1600= 900.

- **Conditions of Profit Maximisation :**
  - **Necessary Condition :** MR must be equal to MC.
  - **Supplementary Condition :** MC should cut MR from below. It simply means after equilibrium point MC should be greater than MR or MC is rising.
- **Producer's Equilibrium :**
  - **Total Revenue – Total Cost Approach**
    The vertical distance between TR and TC is the measure of profit. When this distance is highest, that is the point of equilibrium.
  - **Marginal Revenue – Marginal Cost Approach**
    When MR = MC and MC is rising.
    Break-even point, where TR = TC
- **Firm's Equilibrium under Time Period**
  - **In Short Run :** Three conditions :
    (a) MR = MC
    (b) MC should be greater than MR after equilibrium point.
    (c) Price should be either equal to or more than AVC.
  - **In Long Run**
    (a) MR = MC
    (b) After equilibrium MC should be greater than MR.
    (c) Price should be either equal to or more than AC.
- **Producer :** A producer is one who produces and / or sells goods and services for the generation of income.
- **Normal profits :** Normal profits are defined as the minimum return that the producer expects from his capital invested in the business. It is a situation when TR = TC.
- **Abnormal profits or Supernormal profits :** It is a situation when TR > TC.
- **Sub-normal profits (or losses) :** It is a situation when TR < TC.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. Define producer's equilibrium. [OEB]

Ans. Producer's equilibrium can be defined as when total profits are maximised at the level of output.

Q.2. What is the necessary condition for the producer to be in equilibrium? AI [OEB]

Ans. The necessary condition for the producer to be in equilibrium is that marginal revenue should be equal to marginal cost.

Q.3. The shape of marginal cost is U-shaped, what is the reason for this shape of marginal cost curve? [OEB]

Ans. The shape of marginal cost is U-shaped because marginal cost falls as output is increased and then it rises as output is further increased.

Q.4. At a certain level of output, the Marginal Cost of a firm is above its Marginal Revenue. Can this be its equilibrium output? [OEB]

Ans. No, it cannot be its equilibrium output because the **MC exceeds the MR.** The firm is running in a loss.

Q.5. Is it enough to say that profit is maximised when MC = MR? [OEB]

Ans. No, along with MC = MR, another necessary condition is that **MC should be rising.**

Q.6. State whether the following statement is true or false with the reason: [OEB]

"Under producer equilibrium, MC should cut MR from above."

Ans. False

Reason: The given statement is false because under producer equilibrium, MC should cut MR from below and after that MR=MC, MC should be greater than MR.

Q.7. "Under producer equilibrium, both MR and MC should be equal." [OEB]

Ans. True

Reason: The given statement is true because the necessary condition for the producer to be in equilibrium is that marginal revenue should be equal to marginal cost.

3 Marks Questions

Q.1. Explain the conditions of producer's equilibrium with the help of a schedule, assuming that the producer can sell more only by lowering the price. [OEB]

Ans. Producer's Equilibrium : It refers to a situation in which a producer **maximises his profit** and **minimises his loss**.

Conditions of Producer's Equilibrium:

(i) MR = MC.

(ii) MC > MR after equilibrium.

Price (in ₹)	Output (Units)	TC (₹)	MC (₹)	TR (₹)	MR (₹)
8	1	6	6	8	8
7	2	11	5	14	6
6	3	15	4	18	4
5	4	18	3	20	2
4	5	23	5	20	0

The Producer is in equilibrium at 3 units of output.

Here both the conditions are fulfilled :

(i) MR = MC

(ii) MC > MR after equilibrium.

Q.2. From the following table, find out the level of output at which the producer is in equilibrium. Give reasons for your answer. [OEB]

Output (Units)	Average Revenue (AR in ₹)	Total Cost (TC in ₹)
1	**12**	**14**
2	**12**	**26**
3	**12**	**35**
4	**12**	**52**
5	**12**	**64**
6	**12**	**70**

Concept Applied: Producer's Equilibrium using TC – TR Approach

Ans.

Output (Units)	AR (in ₹)	TC (in ₹)	TR (in ₹)	Profit in ₹ (TR – TC)
1	12	14	12	(–) 2
2	12	26	24	(–) 2
3	12	35	36	+ 1
4	12	52	48	(–) 4
5	12	64	60	(–) 4
6	12	70	72	+2

Producer is in equilibrium at 6th unit of output.

Reasons: The producer is in equilibrium when the difference between the total revenue and total cost (i.e., profit) is maximum. At the sixth unit of output, producer gets maximum profit which is equal to 2 in this case.

Q.3. A producer can sell any quantity of a commodity at a given price of ₹ 10 per unit. His cost schedule is given below: [OEB]

Output (Units)	1	2	3	4	5	6
Total Cost (₹)	12	22	31	38	48	60

How many units should he produce to get maximum profits ? Explain.

Concept Applied: Producer's Equilibrium using MC – MR Approach

Ans.

Output (Units)	MR (₹)	TC (₹)	MC (₹)
1	10	12	12
2	10	22	10
3	10	31	9
4	10	38	7
5	10	48	10
6	10	60	12

The producer is in equilibrium producing 5 units because this meets both the conditions of equilibrium.

(i) MC = MR

(ii) MC > MR after equilibrium.

Q.4. From the following schedule about a producer, find out his profit maximisation output. [OEB]

Output (Units)	1	2	3	4	5	6
Price (Per Unit ₹)	20	20	20	20	20	20
Total Cost (₹)	24	44	62	76	96	120

Ans.

Output	MR (₹)	TC (₹)	MC (₹)
1	20	24	24
2	20	44	20
3	20	62	18
4	20	76	14
5	20	96	20
6	20	120	24

The producer is in equilibrium producing 5 units because this meets both the conditions of equilibrium.

(i) MC = MR

(ii) MC > MR after equilibrium.

Q.5. From the following information about a producer find out the level of output at which he will be in equilibrium. [OEB]

Output (Units)	1	2	3	4	5
Price (Per Unit ₹)	10	9	8	7	6
Marginal Cost (₹)	10	6	3	4	7

Ans.

Output	Price (₹)	TR (₹)	MR (₹)	MC (₹)
1	10	10	10	10
2	9	18	8	6
3	8	24	6	3
4	7	28	4	4
5	6	30	2	7

The producer is in equilibrium producing 4 units because this meets both the conditions of equilibrium.

(i) MC = MR

(ii) MC > MR after equilibrium.

4 Marks Questions

Q.1. Maximum profit implies the equilibrium position of a firm, but every equilibrium does not imply maximum profit of the firm. Justify this statement with the help of a suitable diagram.

[ISC Specimen Paper 2023 – 24]

OR

A producer is in equilibrium when MR = MC. Explain this statement with the help of a diagram. [ISC 2019]

OR

Discuss producer's equilibrium in perfect competition, using MR and MC approach. [ISC 2017]

OR

Explain how a producer can maximise profit by using MR and MC curves. [ISC 2015]

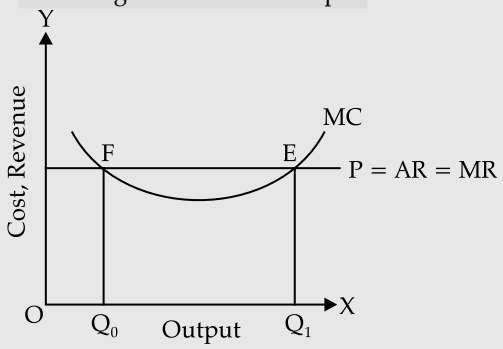
Concept Applied Producer's equilibrium

Ans. Only MR = MC does not ensure maximum profit. A firm earns maximum profit at the equilibrium level of output and has no tendency to change its output.

Equilibrium conditions

Necessary condition: MR = MC

Second condition: The MC curve must intersect the MR curve from below and MC must be increasing at that level of output.



Explanation:

The figure shows MR = MC at two points F and E. **Point F satisfies only the necessary condition, not the sufficient condition.**

∴ The firm cannot be in equilibrium at point F or output OQ_0 where MR = MC.

The firm will increase its output till OQ_1. Between OQ_0 and OQ_1 level of output, the firm can earn profit as MR > MC.

Point E at OQ_1 level of output satisfies both the conditions.

∴ **Point E is the equilibrium point which ensures maximum profit.**

[ISC Marking Scheme, 2019]

Examiner's Comment

- Many candidates did not explain why equilibrium could not be at two points although MR=MC at both the points. Several candidates were unable to explain when MC>MR or MC<MR. Some candidates did not write the conditions of equilibrium. A few candidates did not mention the reason of a firm being in stable condition or at equilibrium.

Answering Tips

- Tell the students to write the two conditions of equilibrium correctly.
- Explain that the MC curve should cut price line at two points.
- Explain about the output where a producer should stop production.
- Ensure that the students understand the economic implications of MR and MC and why second MR=MC point is the point of equilibrium.

Q.2. From the following table identify at which stage the producer is in equilibrium? [OEB]

Units of output	MR	MC
0	10	20
1	10	18

This question is for practice and its solution is available at the end of the chapter

2	10	10
3	10	15
4	10	23

Q.3. Identify from the following table at which stage the producer is in equilibrium? [OEB]

Output	MR	MC
0	15	20
1	15	15
2	15	10
3	15	8
4	15	15
5	15	20

Ans.

Output	MR	MC
0	15	20
1	15	15
2	15	10
3	15	8
4	15	15
5	15	20

The producer is in equilibrium at 4 units of output because here at this point MR = MC and after that point MC becomes greater than MR. Also, at 1 unit of output, MR = MC, but after that MC becomes less than MR and hence this point is not in equilibrium.

Q.4. Give the meaning of producer's equilibrium. A producer produces that quantity of his product at which marginal cost and marginal revenue are equal. Is he earning maximum profit? Give reasons for your answer. [OEB]

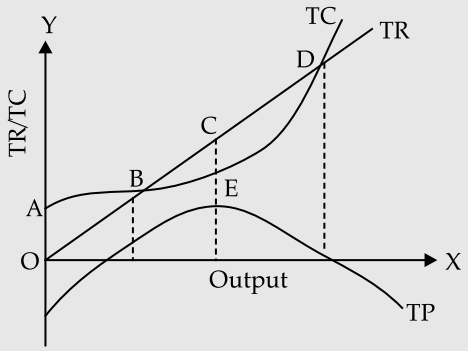
Concept Applied Producer's equilibrium

Ans. Producer's Equilibrium: A producer (firm) is said to be in equilibrium when the firm is producing that quantity of output which gives the firm maximum profit. For a firm, to be in equilibrium, two conditions must be fulfilled. First, and the necessary condition is that firm's marginal cost equals marginal revenue. Second, along with the first condition is that MC must be greater than MR beyond the level of output at which MC = MR. Therefore, fulfilment of the first condition alone does not ensure maximum profits. It is possible that MC = MR condition may be fulfilled at more than one output level but only that output level beyond which MC > MR is the maximum profits output level.

6 Marks Questions

Q.1. Explain how a producer attains equilibrium using the TR and TC approach. [ISC 2018]

Ans.



Profit = TR – TC

Break-even point = normal profit is satisfied at B and D.

- Where intersection of TC + TR (TR covers all TC)
- Profit is maximized on a graph where the vertical distance between TR and TC is the greatest.
- At break-even point B, the industry should PRODUCE MORE.
- At profit maximum point – the vertical distance between TR and TC is at its greatest, meaning the firm is earning maximum economic profits; therefore, industry should keep producing at this level of output, because more profits can still be made.
- At break-even point 2: STOP PRODUCING or REDUCE the quantity being produced. TR and TC curves are equal. Meaning the firm is now only earning a normal profit.
- Distance between TR and TC curves is measured by drawing tangents to both the curves. The distance is maximum when the tangent lines are parallel at point B. This is a situation of maximum profits and producer's equilibrium.
- At point D (second break-even point) all profits are earned.
- Beyond D, only loss.

[ISC Marking Scheme, 2018]

Examiner's Comment

- While explaining how a producer attains equilibrium using TR and TC approach, several candidates incorrectly drew the TC curve starting from the origin. A point of profit maximisation with reasons for the same was not given by many candidates. A few candidates drew the TP curve.

Answering Tips

- The students should draw the TC curve starting from the y-axis explaining the reason for the same.
- Practice how the distance between TR and TC indicates the profit. The students need to draw the tangents to calculate maximum profit.
- Learn to draw the TP curve with the explanation as to why a producer will not produce beyond a point.

Q.2. Giving reasons identify the equilibrium level of output and find profit at this output using Marginal Cost and Marginal Revenue approach from the following: [OEB]

Output (Units)	1	2	3	4	5
Total Revenue (₹)	10	20	30	40	50
Total Cost (₹)	12	22	30	40	52

Ans.

Output (Units)	TR (₹)	TC(₹)	MR (₹)	MC (₹)
1	10	12	10	12
2	20	22	10	10
3	30	30	10	8
4	40	40	10	10
5	50	52	10	12

The producer is in equilibrium at 4 units of output as it fulfils both conditions, which are:

(i) MR = MC

(ii) MC > MR beyond equilibrium.

Profit (π) = TR – TC

= 40 – 40

Profit = 0

Q.3. Give reasons to identify the equilibrium level of output and find profit at this output using Marginal Cost and Marginal Revenue approach from the following: [OEB]

Output (Units)	1	2	3	4	5
Total Cost (₹)	7	13	20	28	37
Total Revenue (₹)	7	14	21	28	35

Ans.

Output (Units)	TC (₹)	TR(₹)	MC (₹)	MR (₹)
1	7	7	7	7
2	13	14	6	7
3	20	21	7	7
4	28	28	8	7
5	37	35	9	7

The producer is in equilibrium at 3 units of output, as it fulfils both conditions, which are:

(i) MR = MC 1

(ii) MC > MR beyond equilibrium 1

Profit (π) = TR – TC

= 21 – 20

or Profit (π) = ₹ 1.

OBJECTIVE TYPE QUESTIONS

[A] Multiple Choice Questions

Q.1. A firm produces its profit maximising level of output only when: [ISC Specimen Paper 2022 – 23]

(A) MR = AR **(B)** MR > MC

(C) AR < MR **(D)** MC = MR

Ans. Option (D) is correct.

Explanation: MR is the addition to total revenue from the sale of one more unit of output and MC is the addition to total cost for increasing the production by one unit. The basic aim of every producer is to maximize the profit. For this, a firm compares its MR with its MC.

Q.2. The shape of TVC is initially concave downwards and subsequently concave upwards because: [ISC Semester-I, 2022]

(A) Initially TVC increases at decreasing rate and then increases at increasing rate.

(B) Initially TVC decreases at decreasing rate and then increases at diminishing rate.

(C) Initially increases at increasing rate and then increases at decreasing rate.

(D) Initially decreases at decreasing rate and then increases at decreasing rate.

Ans. Option (A) is correct.

Explanation: The total variable cost first increases at a decreasing rate and then increases at an increasing rate. This leads to the TVC curve to be concave downwards and then concave upwards.

Q.3. The type of cost that can be entered in the book of accountancy is: [ISC Semester-I, 2022]

(A) Opportunity cost.

(B) Real cost.

(C) Economic Cost.

(D) Money Cost.

Q.4. Which of the following indicates fixed cost? [ISC Semester-I, 2022]

(A) Expenses on raw materials.

(B) Salaries of casual workers.

(C) Electricity bills.

(D) Interest in loans.

This question is for practice and its solution is available at the end of the chapter

Ans. Option (D) is correct.

Explanation: Interest in loans has to be paid at the same rate even if there is no production, thus, it indicates fixed cost.

Q.5. If TC = ₹50 at zero level of output, then what will be the value of TFC? [ISC Semester-I, 2022]

(A) ₹30 **(B)** ₹40
(C) ₹50 **(D)** ₹60

Ans. Option (C) is correct.

Explanation: The value of TFC is equal to the TC at zero level.

We know TC = TFC + TVC. At zero level, TVC = 0. Thus, TC = TFC + 0 or TC = TFC.

Q.6. Match the cost concepts in List I with their definitions in List 2. [ISC Semester-I, 2022]

List I (Cost Concepts)		List II (Description)	
(A)	Explicit Cost	**(i)**	Change in total cost for a unit change in output.
(B)	Marginal Cost	**(ii)**	Value of inputs owned and used in production.
(C)	Implicit Cost	**(iii)**	Costs which a business incurs

(A) (A) – (ii), (B) – (i), (C) – (iii)
(B) (A) – (i), (B) – (ii), (C) – (iii)
(C) (A) – (iii), (B) – (i), (C) – (ii)
(D) (A) – (iii), (B) – (ii), (C) – (i)

Ans. Option (C) is correct.

Explanation: Explicit Costs are those cash payments which firms make to outsiders for their services and goods. For example, the use of a factory building on rent by the producer as a factory outlet for production of goods.

Implicit cost are the costs which the producer incurs by the use of the services of the factor of production which are owned by the producer himself. For example, the use of own building as a factory by the producer.

Marginal Cost is the cost incurred in the production of one more unit of output. It is the additional cost incurred for the increase in the production by one more unit.

Q.7. Which of the following statements are true? [ISC Semester-I, 2022]

(A) MC is unaffected by variable cost.
(B) MC is unaffected by fixed cost.
(C) MC equals change in TR.
(D) MC is unaffected by price.

Ans. Option (B) is correct.

Explanation: Marginal Cost is the change in the total cost. When we see Total cost is nothing but the summation of Total Fixed Cost and Total Variable Cost. As Total fixed cost is constant at all levels of production, Marginal cost will never be affected by the fixed cost but is affected by the change in the variable cost. Marginal cost has no relation with total revenue and price of the product.

Q.8. When MR is perfectly elastic, TR is: [ISC Semester-I, 2022]

(A) Upward sloping. **(B)** Downward sloping.
(C) Concave to origin. **(D)** Convex to origin.

Ans. Option (A) is correct.

Explanation: MR is perfectly elastic in the case of perfect competition market. Here the AR curve is also perfectly elastic and is equal to the MR and Price. In such a case when the AR is constant, the TR curve will increase at the same rate making it upward sloping.

Q.9. Cost of exertion, pains and sacrifices is known as: [ISC Specimen Paper, Semester-I, 2021-22]

(A) Opportunity costs **(B)** Production costs
(C) Real costs **(D)** Explicit Costs

Ans. Option (C) is correct.

Explanation: Real cost is the qualitative sum of all the efforts made by the owners of factor of production.

Q.10. What is the nature of the expenditure on raw material, power and labour? [ISC Specimen Paper, Semester-I, 2021-22]

(A) Fixed **(B)** Variable
(C) Production costs **(D)** Explicit

Ans. Option (B) is correct.

Explanation: The expenditure on raw materials, power and labour is affected by the number of output produced, thus, they are variable in nature.

Q.11. What is the correct formula for calculating the Total variable cost? [ISC Specimen Paper, Semester-I, 2021-22]

(A) TVC = ∑MC **(B)** TVC = TC – TFC
(C) TVC = AVC x Q **(D)** All

Q.12. Which of the following statements is true? [ISC Specimen Paper, Semester-I, 2021-22]

(A) $AR = \frac{MR}{Q}$ **(B)** TR= MR x Q
(C) TR = ∑MR **(D)** All

Ans. Option (C) is correct.

Q.13. A competitive firm will be in equilibrium when: [ISC Specimen Paper, Semester-I, 2021-22]

(A) MR= MC and MC cuts MR from below
(B) AR = AC; AC cuts AR from above
(C) TR= TC **(D)** None of these

Ans. Option (A) is correct.

Explanation: The two conditions for producer's equilibrium are when the Marginal Revenue (MR) is equal to the Marginal Cost (MC) and marginal cost curve cuts the marginal revenue curve from below.

Q.14. AVC and AC curves cannot intersect each other because: [ISC Specimen Paper, Semester-I, 2021-22]

(A) MC starts falling.
(B) TC- TVC =TFC which is constant.
(C) TC does not increase.
(D) None of the above.

Ans. Option (D) is correct.

Explanation: AVC and AC curve tend to intersect but never intersect each other as the value of AFC tends to become zero but never becomes zero.

This question is for practice and its solution is available at the end of the chapter

Q.15. What is the reason for the U shape of short run average cost curves?

[ISC Specimen Paper, Semester-I, 2021-22]

(A) Law of Demand
(B) Law of DMU
(C) Law of variable proportions
(D) Law of supply

Ans. Option (C) is correct.

Explanation: The law of variable proportions states that when more and more variable factors are employed the total product first increases at an increasing rate and then increases at a decreasing rate. If we apply this law with average costs, the per unit cost first decreases as the output increases at an increasing rate and then afterwards starts increasing as the output increases at a decreasing rate. This gives the U shape to the short run average cost curves.

Q.16. Opportunity cost of a good is the _____.

[ISC Specimen Paper, Semester-I, 2021-22]

(A) Anticipated cost **(B)** Next best alternative
(C) Upcoming cost **(D)** Real cost

Q.17. Social cost is:

[ISC Specimen Paper, Semester-I, 2021-22]

(A) Implicit cost + Explicit cost
(B) Explicit cost+ Opportunity cost
(C) Private cost + External cost
(D) Private cost + Real cost

Ans. Option (C) is correct.

[B] Assertion Reason Questions

Directions: In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.1. Assertion (A): When Average Cost is constant, AC curve is at its minimum point.

Reason (R): At this point, MC curve cuts AC curve, which implies MC = AC. **[OEB]**

Ans. Option (B) is correct.

Explanation: At the minimum point of the AC, the MC cuts the AC curve and average cost is constant because the TR is increasing at a constant rate.

Q.2. Assertion (A): The difference between AC and AVC is due to AFC.

Reason (R): As output increases AFC decreases, so the difference between AC and AVC decreases.

[OEB]

Q.3. Assertion (A): A firm is able to sell more quantity of a good only by lowering the price.

Reason (R): The firm's Marginal Revenue, as it goes on selling, would be less than Average revenue. **[OEB]**

Ans. Option (B)is correct.

Explanation: When a firm can sell more only by lowering the price it is possible that marginal revenue becomes negative after a level of output.

Q.4. Assertion (A): When MR is positive, TR tends to increase.

Reason (R): As under monopoly, more of the commodity can be sold by lowering down the price of the commodity. Hence AR curve is downward sloping from left to right. **[OEB]**

Ans. Option (B) is correct.

Explanation: When MR is positive, TR tends to increase as there is an increase in the TR either in increasing rate or decreasing rate and MR is just the change in TR with one extra unit of output sold.

Q.5. Assertion (A): After short-term equilibrium level of output, the Marginal Cost of a firm is above its Marginal Revenue.

Reason (R): At equilibrium level of output, the Marginal Cost is equal to the Marginal Revenue. **[OEB]**

Ans. Option (A) is correct.

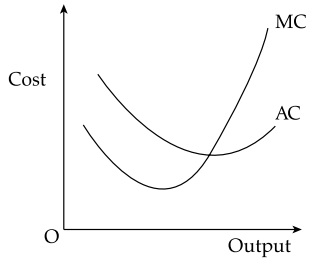
Explanation: At equilibrium level of output, MC is equal to MR as the firm earns normal profit at that point and thus after equilibrium the MC is above MR.

Case Based Questions

[A] Multiple Choice Questions

I. Observe the following diagram:

[ISC Semester-I, 2022]



(a) Choose the correct option:

(A) When MC > AC, AC falls.
(B) When MC < AC, AC rises.
(C) When MC = AC, AC falls.
(D) When MC > AC, AC rises.

Ans. Option (D) is correct.

(b) MC cuts AC at AC's _____ point.

(A) Maximum point.
(B) Minimum Point.
(C) When AC is rising.
(D) When AC becomes parallel to X axis.

Ans. Option (B) is correct.

This question is for practice and its solution is available at the end of the chapter

II. Choose the correct answer to fill the following table: [ISC Semester-I, 2022]

Output	Price	TR	AR
2	9		
3	7		

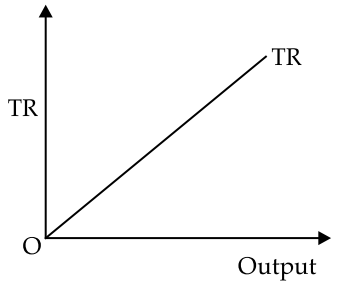
(a) The TR will be:
(A) 18, 21 **(B)** 20, 27
(C) 15, 16 **(D)** 20, 21

Ans. Option (A) is correct.
Explanation: TR at 2 unit = 9 × 2 = 18; TR at 3 units = 7 × 3 = 21

(b) The AR will be:
(A) 9, 8 **(B)** 9, 7
(C) 10, 8 **(D)** 7, 9

Ans. Option (B) is correct.
Explanation: AR is equal to the price.

III.



(a) The TR curve in the above diagram belongs to:
(A) Monopoly **[OEB]**
(B) Monopolistic Competition
(C) Oligopoly
(D) Perfect Competition

Ans. Option (D) is correct.
Explanation: Total Revenue Curve is an upward rising straight line curve starting from the origin. Price is constant in nature in perfect market.

(b) The curve is a straight line because:
(A) Price is increasing with output.
(B) Price is decreasing with output.
(C) Price remains constant.
(D) Marginal Revenue is increasing.

Ans. Option (C) is correct.
Explanation: TR increases in same proportion as increase in output sold due to the Price remaining constant.

(c) TR is calculated as:
(A) Price × Output **(B)** AR × MR
(C) AC × MR **(D)** TC × Price

Ans. Option (A) is correct.

IV. Read the following passage and answer the questions that follows:

Jordan Cement Factories Company was set up in December 1951 as a share holding company. In March 1954, the company commenced business with the first bag of cement.

In order to ascertain the cost of products for a particular period of time, the company prepares cost sheet, the cost sheet data are collected from various statements of accounts which have been written in cost accounts either on day to day or regular records. The main elements of cost sheet are prime cost, work cost and cost of production.

The main principle that underlines the cost classifications of main elements of the cost is fixed and variable cost basis. The company does not consider any others basis like direct and indirect costs or revenue and capital cost or functional classification for cost classification. Fixed and variable cost is based on the changes in activity or volume. Fixed cost or period cost remain unchanged in spite of changes in volume or activity.

Variable cost or product cost vary in complete proportion to the volume of output. Capital and revenue basis depends on the purpose of expenditure. Any cost incurred in purchasing assets either to earn income or increasing the earning capacity of the business is known as capital cost. But any cost incurred for the purpose of maintaining the earning capacity of the business it is revenue expenditure. **[OEB]**

(a) For what reason is the cost sheet prepared?

Ans. The cost sheet is prepared by the company to ascertain the cost of the product at a particular time.

(b) What is the main principle underlying the cost classification?

Ans. The main principle underlying the cost classification is the main element of the cost in fixed cost and variable cost basis.

(c) "Variable cost or product cost vary in complete proportion to the volume of output." Explain.

Ans. Variable cost or product cost vary in complete proportion to the volume of output because variable cost varies in proportion to the volume of output. In other words, variable cost first increases at a decreasing rate and then increases at an increasing rate as the output produced increases.

(d) Classify the following expenditures as fixed or variable:
(A) Purchase of Assets.
(B) Payment made to labour.
(C) Installation of new machinery.
(D) Payment of interest on loan.

Ans. (A) Fixed Cost.
(B) Variable Cost.
(C) Fixed Cost.
(D) Fixed Cost.

(E) What is the relation between TFC, TVC and TC. (2)

Ans. Total fixed cost curve is parallel to x-axis. Total variable cost curve and total cost curves are inverse S-shaped. Total cost curve and total variable cost curve both first increase at increasing rate and then at decreasing rate.

[B] Subjective Based Questions

V. Read the following passage and answer the question that follows:

The slope of a total revenue curve is particularly important. It equals the change in the vertical axis (total revenue) divided by the change in the horizontal axis (quantity) between any two points. The slope measures the rate at which total revenue increases as output increases. We can think of it as the increase in total revenue associated with a 1-unit increase in output. The increase in total revenue from a 1-unit increase in quantity is marginal revenue. Thus, marginal revenue (MR) equals the slope of the total revenue curve.

How much additional revenue does a radish producer gain from selling one more pound of radishes? The answer, of course, is the market price for 1 pound. Marginal revenue equals the market price. Because the market price is not affected by the output choice of a single firm, the marginal revenue the firm gains by producing one more unit is always the market price. The marginal revenue

curve shows the relationship between marginal revenue and the quantity a firm produces. For a perfectly competitive firm, the marginal revenue curve is a horizontal line at the market price. If the market price of a pound of radishes is $0.40, then the marginal revenue is $0.40. Marginal revenue curves for prices of $0.20, $0.40, and $0.60. In perfect competition, a firm's marginal revenue curve is a horizontal line at the market price.

Price also equals average revenue, which is total revenue divided by quantity. To obtain average revenue (AR), we divide total revenue by quantity, Q. Because total revenue equals price (P) times quantity (Q), dividing by quantity leaves us with price. **[OEB]**

—Source: Price, Marginal Revenue, and Average Revenue – Open Textbook for Hong Kong.

(a) What does the slope of Total Revenue equal?

Ans. The slope of Total Revenue curve equals the marginal revenue.

(b) What does the Marginal Revenue equal? Why?

Ans. The Marginal revenue equals the Market Price because a single firm does not affect it.

(c) What is the relationship between Marginal Revenue and Total Revenue in case of an imperfect competition firm?

Ans. Relationship between Total Revenue and Marginal Revenue:

(i) Initially or at first unit TR = MR

(ii) When TR increases at increasing rate then MR also increases.

(iii) When TR increases at constant rate MR is also constant.

(iv) When TR increases at a diminishing rate then MR declines.

(v) When TR is maximum, MR is zero.

(vi) When TR declines MR is negative.

(d) Explain the shape of the revenue curves in imperfect competition market.

Ans. The Total Revenue curve under monopoly and monopolistic competition is upward rising at the start and then after reaching the maximum starts to fall. This is because the TR first increases at a diminishing rate and after certain level (maximum) of output the TR starts to decrease.

In monopoly and monopolistic competition, AR and MR both are downward sloping and MR is always below AR. The main difference in these two markets is that in monopolistic competition AR and MR curve is more elastic than monopoly. Under monopoly and monopolistic competition, more of the commodity can be sold only at a lower price. This implies an inverse relationship between price of the commodity and demand for the firm's output. Hence, a downward sloping firm's demand curve. Marginal revenue also falls but the rate of fall in MR is greater than in AR. This is due to the fact that not only the additional units will be sold at lowered price but the whole units are to be sold at lower price.

Negative (MR) is possible only when price is declining under monopoly or monopolistic competition. It is not possible in case of perfect competition, where price remains constant for a firm.

(e) Explain the shapes of the revenue curves in perfect competition market.

Ans. Total Revenue Curve is an upward rising straight line curve starting from the origin. Price is constant in nature in perfects market. TR increases in same proportion as increase in output sold due to the Price remaining constant.

Average Revenue Curve is a horizontal line parallel to the X – axis. Marginal Revenue is equal to the Average Revenue in the case of Perfect Competition Market. This is because, under perfect competition, a firm is a price taker and cannot change price. The price remains constant and as a result AR and MR also remain constant and at the same level as the price.

Solutions for Practice Questions (Topic-1)

1 Mark Questions

5. Average fixed cost is calculated by dividing total fixed cost by output whereas Average Variable cost can be defined as total variable cost divided by output.

2 Marks Questions

5. **Meaning of social cost:** Social cost refers to the cost that the society has to bear on account of the production of a commodity. It includes private cost +external cost. External cost is what the society has to bear in terms of pollution, smoke, noise pollution etc.

For example: the cost of purifying contaminated water by the society.

[ISC Marking Scheme, 2016]

Examiner's Comment

- A number of candidates did not write social cost = private cost + external cost. They explained social cost as external cost – pollution, as example. They ignored private cost.

Answering Tip

- Learn the concept of 'Social' cost with different examples. Inclusion of private cost in social cost should be explained.

3 Marks Questions

5 Money cost refers to money expenses which the firm has to incur in purchasing or hiring the factor services.

Real Cost refers to the efforts and sacrifices made by the owners of factors of production used in the production of a commodity.

[ISC Marking Scheme, 2017]

Examiner's Comment

- While explaining 'real cost', several candidates did not mention 'efforts and sacrifices'. A large number of candidates explained 'money cost' correctly.

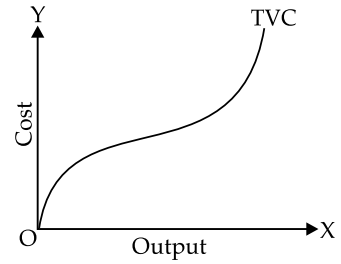
Answering Tip

- Explain the concepts of different costs with the help of examples.

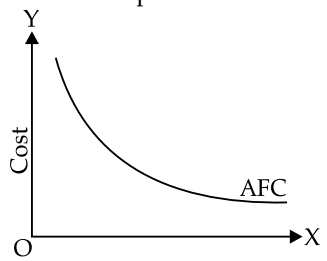
4 Marks Questions

2. Cost can be defined as the total expenditure made by the firms in producing output. Cost is sum total of fixed cost and variable cost.

(a) Total variable Cost: It can be defined as expenditure incurred on various variable factors of production. For example: cost incurred on raw material, wages paid to casual labourers etc.



(b) Average fixed cost: It can be defined as total fixed cost divided by output. Average fixed cost is per unit of fixed cost of production.



6 Marks Questions

3. Average Cost : Cost per unit of output is called Average Cost. It is obtained by dividing the total cost by the quantity of output.

$$\mathbf{AC} = \frac{TC}{Q}$$

Marginal Cost : Marginal cost is the addition made to the total cost by the production of one more unit of a commodity.

$$MC = TC_n - TC_{n-1}$$

$$MC = TVC_n - TVC_{n-1}$$

$$\text{or,} \quad MC = \frac{\Delta TC}{\Delta Q} = \frac{\Delta TVC}{\Delta Q}$$

Relationship between Marginal Cost and Average Cost:

Both MC and AC are derived from the total cost. They bear unique relationship.

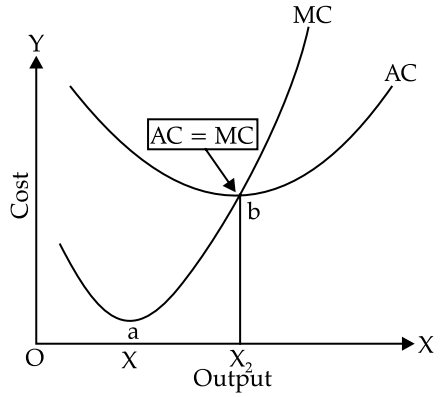
The relationship between MC and AC can be stated as under:

(a) When AC falls with increase in output, MC is lower than AC, i.e., MC curve lies below the AC curve. However, it is not necessary that MC should fall throughout this stage. Actually, MC rises earlier than AC.

(b) When AC rises with increase in output, MC is higher than AC, i.e., MC curve lies above the AC curve.

(c) At the level of optimum output, average cost is minimum and constant. Here, MC stands equal to AC, i.e., MC pulls AC horizontally.

(d) At zero level of output both AC and MC are indeterminate.



Solutions for Practice Questions (Topic-2)

1 Mark Questions

2. Average revenue can be defined as the total revenue divided by no. of units sold.

2 Marks Questions

2. Market is perfect competition. AR = MR as the price is same or uniform

OR

Perfectly competitive firm is price taker.

[ISC Marking Scheme 2015]

Examiner's Comment

- Identification of perfect competition was correct in most cases. However, several candidates were not able to give the reason for equality between AR and MR

Answering Tip

- Students should identify different forms of market on the basis of important characteristics.

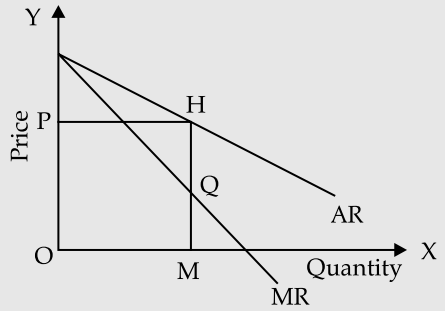
3 Marks Questions

2. The relationship between AR and MR of a firm under imperfect competition:

A firm under imperfect competition (Monopoly, Monopolistic) is required to reduce the price if it wants to sell more units of its product.

(i) Marginal revenue is always less than the average revenue or price.

(ii) The rate of fall of MR will be twice as much as the rate of fall of AR curve.



[ISC Marking Scheme, 2019]

Examiner's Comment

- Most of the candidates wrote the different types of relations between AR and MR except the required one. Several candidates wrote AP and MP in place of AR and MR. Many candidates did not show the MR curve going down the axis.

Answering Tip

- Practice with the help of a diagram the relationship between AR and MR of a firm.

6 Marks Questions

2. AR and MR curves under Perfect Competition: See Answer to Question No. 3 of 3-mark Questions.

AR and MR curves under Imperfect Competition: Under imperfect competition, a producer can increase sales by decreasing the price. As a result, each additional unit is sold at the price lower than the price earned for the previous unit. This is the reason MR curve is below AR curve under imperfect competition.

Solutions for Practice Questions (Topic-3)

4 Marks Questions

2.

Units of Output	MR	MC
0	10	20
1	10	18
2	10	10
3	10	15
4	10	23

The producer is in equilibrium at 2 units of output because there marginal revenue (10) is equal to marginal cost (10), and after that point Marginal cost (15) should be greater than marginal revenue(10).

Solutions for Objective Type Questions

[A] Multiple Choice Questions

3. Option (D) is correct.

Explanation: The money cost of producing a certain output of a commodity is the sum of all the payments to the factors of production engaged in the production of that commodity.

11. Option (D) is correct.

16. Option (C) is correct.

[B] Assertion Reason Questions

2. Option (A) is correct.

Explanation: AFC falls when output is increased. Since, the Total Fixed Cost remains the same with changes in output. Therefore, AFC falls steadily with increase in output. the difference between AC and AVC is due to AFC. As output increase AFC falls, so the difference between AC and AVC falls.

REFLECTION

(a) Can you draw the cost curves, tell why they are shaped that way and calculate them?

(b) Can you tell the different types of costs?

(c) Can you explain the various revenue curves in both perfect and imperfect competition market?

(d) Can you explain how a producer can attain equilibrium?